

Unit: II

Lecture: 4

Memory Management

In order to implement any data structure on an actual computer, we need to use computer memory.

Computer memory is simply a sequence of memory words, each of which usually consists of 4, 8, or 16 bytes (depending on the computer). These memory words are numbered from 0 to $N - 1$, where N is the number of memory words available to the computer. The number associated with each memory word is known as its address. Thus, the memory in a computer can be viewed as basically one giant array of memory words.

Using this memory to construct data structures (and run programs) requires that we manage the computer's memory to provide the space needed for data—including variables, nodes, pointers, arrays, and character strings—and the programs the computer runs.

Ways of keeping track of memory usage:

There are two ways to keep track of memory which is currently in use by variables and which are not:

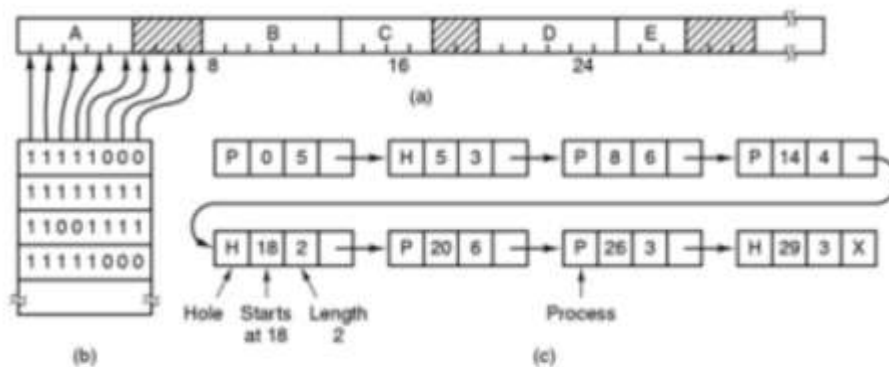


Figure 1: Memory management with bitmap and linked list

- **Memory management with bitmaps:** With a bitmap, memory is divided up into allocation units, perhaps as small as few words and perhaps as large as several kilobytes.
 - Corresponding to each allocation units is a bit in bitmap.
 - The bit is set to zero if the unit is free and 1 if it is occupied.
 - Fig (1), shows part of the memory in part (a) and corresponding bitmap in part(b).

- The size of the allocation unit is important design issue. The smaller the allocation unit, the larger the bitmap.
- **Memory management with link list:** Another way of keeping track of memory is to maintain a linked list of allocated and free memory segments.
 - The linked list is divided into four parts as shown below:

Type of block	Starting address	Length	Pointer to next block
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- The type of block can be a Hole(H) or a process (P), starting address will be the address at which it starts, length signifies the allocation units it occupies and pointer to the next block.
- Fig (1) above shows part of the memory in part(a) and the corresponding linked list in part(c).

Size of the objects:

When there is limited memory resource available, those resources need to be shared among several computer system. But users are not aware of this fact. There are many issues that CPU has to consider during memory management. Memory management often depends upon the size of the object/file. The object/file can be of fixed size or of variable size.

The two examples of systems which used fixed type object are:

- **LISP:** It uses cell to store data which are of same size. For instance for linked list, one cell contains the data and other contains the pointer to the next cell. In binary tree one cell contains the data and other two contains the pointer to the left and the right cell.
- **UNIX:** In this file system, there are fixed sized block in which files can be stored. When file gets deleted from the block, the space can be used by the other file.

The two example of variable sized objects are:

- **Multiprogramming:** contains programs and files of different sizes. Different sized files get different space in main memory which memory management has to take care of.
- **Snobol:** There is a large space of memory available for the objects called heap. The objects are of variable size and takes different storage space in heap.